

SUBHADEEP KOLEY

PhD Student, Computer Science and Engineering, Lehigh University

svk324@lehigh.edu | subhadeepk.github.io | [Google Scholar](https://scholar.google.com/citations?user=svk324) | github.com/subhadeepk | [LinkedIn](https://www.linkedin.com/in/subhadeepk)

Research interests: aerial robotics, physical human-robot interaction, motion planning, learning-based control.
Seeking research internships for Summer 2027.

EDUCATION

Lehigh University

PhD, Computer Science and Engineering. Advisor: Prof. David Saldaña

Aug 2024 – Present

Bethlehem, PA

Indian Institute of Engineering Science and Technology (IIST), Shibpur

B.Tech., Mechanical Engineering. CGPA: 8.56/10

2020 – 2024

Howrah, India

PUBLICATIONS

Under review

- **Subhadeep Koley**, Benjamin Greenberg, Yifei Simon Shao, Juan Aceros, Nadia Figueroa, David Saldaña. **HINT-Blimp: Human INTent Inference from Multimodal Cues for Robotic Blimps**. *Under review, ICRA 2027*. [[arXiv](#)] [[video](#)]
- **Subhadeep Koley**, Subhrajit Bhattacharya, David Saldaña. **Planning Trajectories that Bounce: Reflection Classes for Collision-Tolerant Robots**. *Under review, ICRA 2027*. [[arXiv](#)] [[video](#)]

Conference papers

- Yizhuo Wang, Yuhong Cao, Jimmy Chiun, **Subhadeep Koley**, Mandy Pham, Guillaume Adrien Sartoretti. **ViPER: Visibility-based Pursuit-Evasion via Reinforcement Learning**. *Conference on Robot Learning (CoRL), 2024*. [[paper](#)] [[video](#)] [[code](#)]
- Ajit Das, Saumendra Nath Mishra, **Subhadeep Koley**, Sourav Sarkar, Achintya Mukhopadhyay, Swarnendu Sen. **Early Detection of Thermal Runaway of Lithium-ion Battery: An Experimental Study**. *IEEE Conf. on Sustainable Energy and Future Electric Transportation (SEFET), 2023*. **Best Paper Award**. [[paper](#)]

Workshop papers

- **Subhadeep Koley**, Benjamin Greenberg, Yifei Simon Shao, Nadia Figueroa, David Saldaña. **HINT-Blimp: Human INTent Inference from Multimodal Cues for Robotic Blimps**. *IROS 2026 Workshop on Nonverbal Cues for Human-Robot Cooperative Intelligence, Pittsburgh, 2026*. *Spotlight talk and poster*. [[paper](#)] [[video](#)] [[poster](#)]

Posters and demos

- **HINT-Blimp**: Poster presentation, *Northeast Robotics Colloquium (NERC)*, Princeton University, Oct 2026.
- Live demonstration of [MochiSwarm](#), an open-source swarm of lightweight robotic blimps. *ICRA 2025*, Atlanta.

RESEARCH EXPERIENCE

Graduate Researcher, Swarms Lab, Lehigh University

Advisor: Prof. David Saldaña

Aug 2024 – Present

Bethlehem, PA

- Working on **autonomous navigation and mapping** of large unknown indoor environments using a Unitree Go2.
- Built a **multimodal intent-inference** system for robotic blimps using physical pushes, voice and pointing; combining pushes and voice identified the intended goal **86%** of the time, usually within two interactions, and let users steer the blimp around obstacles only they knew about.

- Developed a planner that lets **collision-tolerant robots** bounce off walls deliberately, grouping paths into reflection classes by wall-contact sequence; the best bouncing path reduced control effort by a median **16.6%** versus a collision-free path in simulation.
- **Won first place** at [Defend the Republic 2024](#) against teams including Georgia Tech, UMich and ASU; built and maintained **16+ robots** and added ultrasonic wall avoidance to the hardware and software stack.

Research Intern, MARMoT Lab, National University of Singapore

Jan 2024 – May 2024

Mentor: Dr. Guillaume Sartoretti

Remote

- Worked on **reinforcement learning** for multi-robot visibility-based pursuit-evasion in unknown environments; published at **CoRL 2024** (ViPER).

SURGE Research Intern, SPiN Lab, IIT Kanpur

May 2023 – Aug 2023

Mentor: Dr. Ketan Rajawat

Kanpur, India

- Part of a 10-person team building ROS software for a **decentralized drone swarm**, deployed on five real drones [[report](#)] [[video](#)].
- Developed a **mission framework** supporting leader-follower flight and independent per-drone missions [[video](#)].
- Built **decentralized leader election** so the swarm elects a new leader on leader failure.
- Developed **inter-drone collision avoidance**, tested under motion capture, and network monitoring and **congestion control** for the swarm's mesh network.

Research Intern, Jadavpur University

Jul 2022 – Dec 2022

Mentor: Dr. Sourav Sarkar

Kolkata, India

- Built a heat- and gas-sensor setup for **early detection of lithium-ion battery thermal runaway**, leading to a Best Paper Award at IEEE SEFET 2023.
- Built a multi-point wireless temperature acquisition system for a Tata Steel-funded project studying cooling rates in rolling-mill cooling beds.

Flight Software Intern, EndureAir Systems

Dec 2022 – Feb 2023

Mentor: Dr. Kuldeep Kumar Dhiman

Noida, India

- Worked with the **PX4** flight stack; built a MATLAB range-extension tool for a drop tank with delayed wing deployment.

AWARDS

- [First place](#), Defend the Republic drone competition, Indiana University Nov 2024
- Best Paper Award, IEEE SEFET Aug 2023

TEACHING

- Teaching Assistant, CSE 340: Design and Analysis of Algorithms, Lehigh University

TECHNICAL SKILLS

Languages	Python, MATLAB
Robotics	ROS, PX4, Gazebo, CoppeliaSim, MoveIt, motion capture
ML / Vision	PyTorch, OpenCV
Hardware	ESP32, ATmega328, Raspberry Pi, SolidWorks, 3D printing